

ECE 222 In-Class Problems 3

In this discussion, we will take a detailed look at some homework problems together to get practice with setting up the problems and equations to find electric potential. This will also reinforce some concepts related to Coulomb's and Gauss's laws.

At the end of class, take a photograph(s) or screenshot(s) of your answers and upload to today's in-class problem assignment on Canvas.

Book problem 4.33:

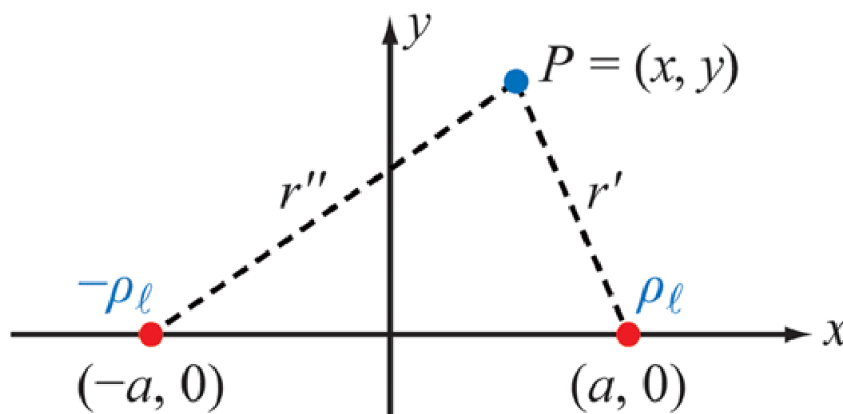
4.33 Show that the electric potential difference V_{12} between two points in air at radial distances r_1 and r_2 from an infinite line of charge with density ρ_ℓ along the z axis is $V_{12} = (\rho_\ell / 2\pi\epsilon_0) \ln(r_2/r_1)$.

- a) You know that you will first need the electric field from an infinite line of charge with density ρ_ℓ . You could just look up the answer in the summary of chapter 4 in the textbook ($\vec{E} = \hat{r} \frac{\rho_\ell}{2\pi\epsilon} \frac{1}{r}$), but can you obtain this result quickly using Gauss's law? You should be able to do this with a diagram and 2 to 4 lines of math.
- b) Calculate the potential difference between the two points.
- c) What would be the most convenient reference point for this problem? Does it actually matter where you choose your reference?

Book problem 4.37:

- *4.37 As shown in Fig. P4.37, two infinite lines of charge, both parallel to the z axis, lie in the x - z plane: one with density ρ_ℓ and located at $x = a$ and the other with density $-\rho_\ell$ and located at $x = -a$. Obtain an expression for the electric potential $V(x, y)$ at a point $P = (x, y)$ relative to the potential at the origin.

Figure P4.37 Problem 4.37.



- a) Just like with problem 4.19, it is easiest to start with the equation for the potential from a single wire at the origin in cylindrical coordinates, and then transform that equation to a shared coordinate system with a shared reference. You already have the equation for the electric potential from an infinite wire from the previous problem, shown below. All I have done here is rename r_2 to r_{ref} .

$$V(r) - V(r_{ref}) = -\frac{\rho_\ell}{2\pi\epsilon} \ln\left(\frac{r}{r_{ref}}\right)$$

The problem tells you to pick the origin as the shared reference point. How would you write r (at point P) and r_{ref} (at origin) in Cartesian coordinates when writing down an equation for the potential from the left wire? What about the right wire?

Keep in mind that an arbitrary distance r from a wire located at (x_{wire}, y_{wire}) would be written as $r = |\vec{R} - \vec{R}_{wire}| = \sqrt{(x - x_{wire})^2 + (y - y_{wire})^2}$

- b) Using the above expressions, write the expressions for the potential from the left and right wires at point P .
- c) Write an expression for the total potential at point P .